

## Scalability & Future Plan

|                      |                                            |
|----------------------|--------------------------------------------|
| <b>Date</b>          | 2 july 2026                                |
| <b>Team ID</b>       | XXXXXX                                     |
| <b>Project Name</b>  | Value-Based Concept Understanding Analyser |
| <b>Maximum Marks</b> | 1 Mark                                     |

### Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

### Current System Limitations:

| S.No | Limitation                                                    | Impact                                                      | Priority to Address |
|------|---------------------------------------------------------------|-------------------------------------------------------------|---------------------|
| 1    | Uses SQLite database for development.                         | Limited scalability for large datasets and concurrent users | High                |
| 2    | Whisper model processing is computationally intensive         | Longer processing time for large audio files                | High                |
| 3    | Application is currently deployed only in a local environment | Remote users cannot access the system                       | High                |

### Scalability Plan:

| S.No | Scalability Aspect | Current State                   | Proposed Upgrade / Solution                                                       |
|------|--------------------|---------------------------------|-----------------------------------------------------------------------------------|
| 1    | User Load          | Supports small number of users. | Deploy on Streamlit Cloud, Render, or Azure with load balancing                   |
| 2    | Data Storage       | SQLite database.                | Upgrade to PostgreSQL or MySQL for better scalability                             |
| 3    | Performance        | Local execution.                | Optimize model inference, caching, and GPU acceleration                           |
| 4    | Security           | Basic user authentication       | Implement HTTPS, Multi-Factor Authentication (MFA), and role-based access control |

### Future Roadmap:

| Phase   | Planned Feature / Enhancement                                  | Target Timeline | Expected Impact                                           |
|---------|----------------------------------------------------------------|-----------------|-----------------------------------------------------------|
| Phase 1 | Cloud deployment on Streamlit Cloud, Render, or Azure          | 1 week          | Improve accessibility and support more concurrent users   |
| Phase 2 | Real-time speech evaluation and instant feedback               | 2 week          | Faster evaluation and improved user experience            |
| Phase 3 | Multilingual speech transcription and concept evaluation       | 3 week          | Support learners speaking different languages             |
| Phase 4 | Personalized AI learning recommendations and progress tracking | 4 week          | Adaptive learning with customized improvement suggestions |